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(54) **PASSIVE DILATION IN CATHETER
INSERTION SYSTEMS**

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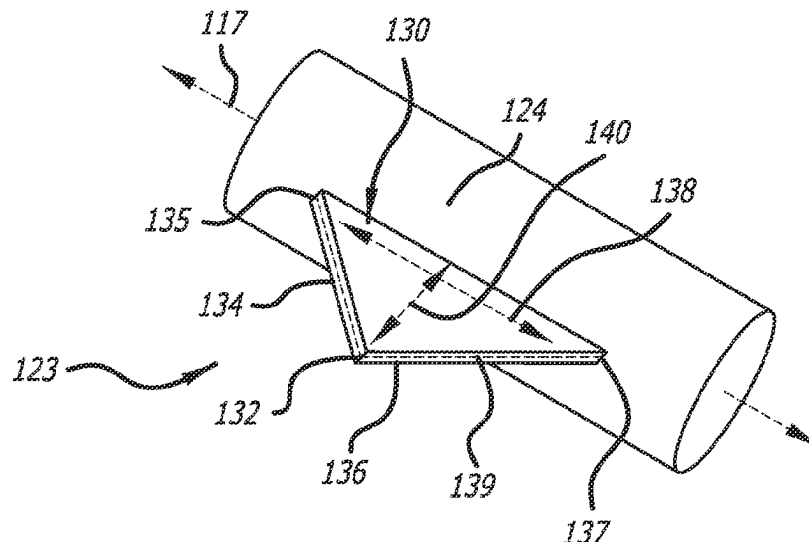
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ABSTRACT

Disclosed herein is a catheter, which in some embodiments
includes a distal section configured to enter a skin insertion
site. The distal section can include a tapered junction having
one or more dilation structures configured to dilate the skin
insertion site. The distal section can further include a distal
portion extending from a distal end of the tapered junction,
the distal portion having a diameter smaller than a proximal
portion of the catheter. In combination, the tapered junction,
the one or more dilation structures, and the specific actions
of the user urging the catheter into the insertion site can
result in an improved fit of the catheter in the insertion site.

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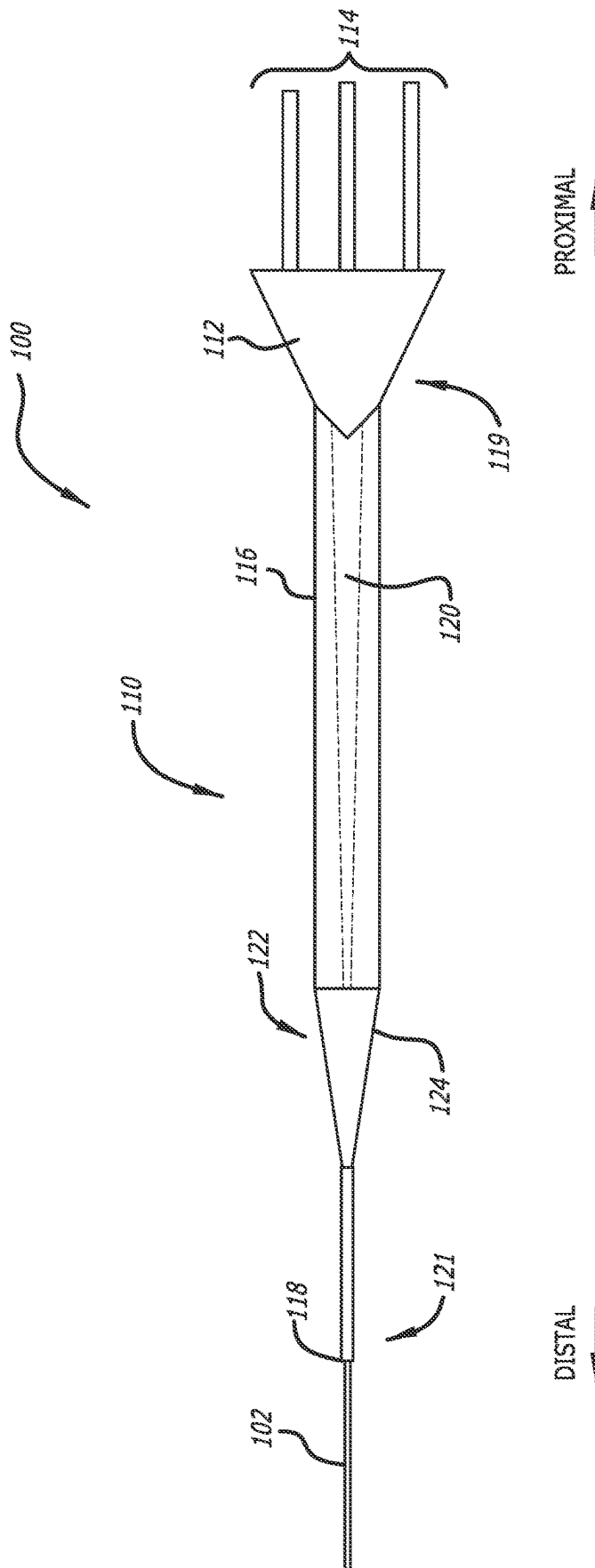
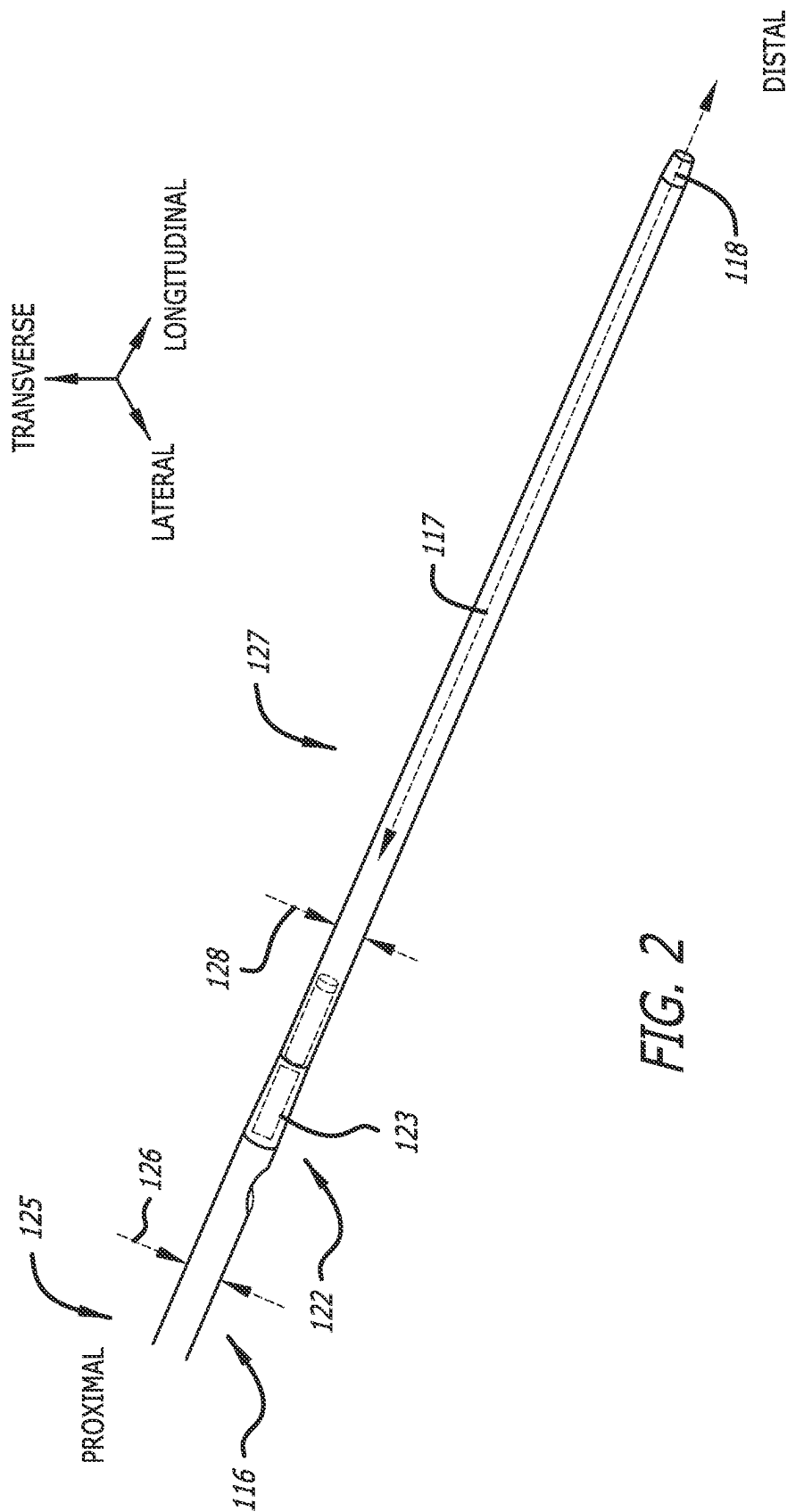
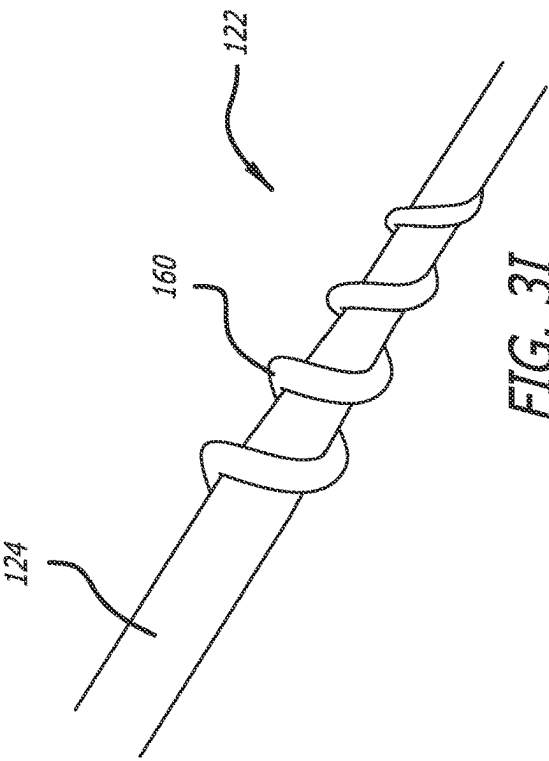
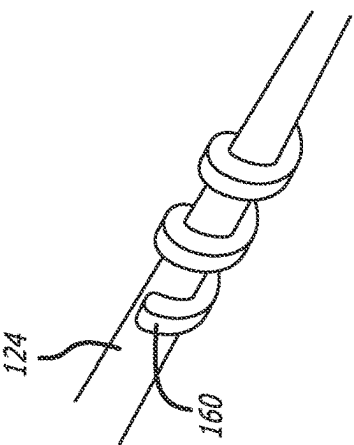
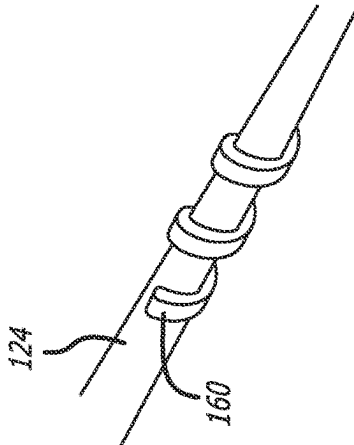
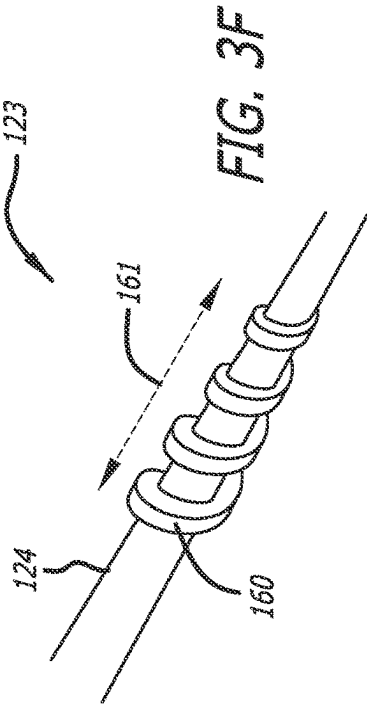
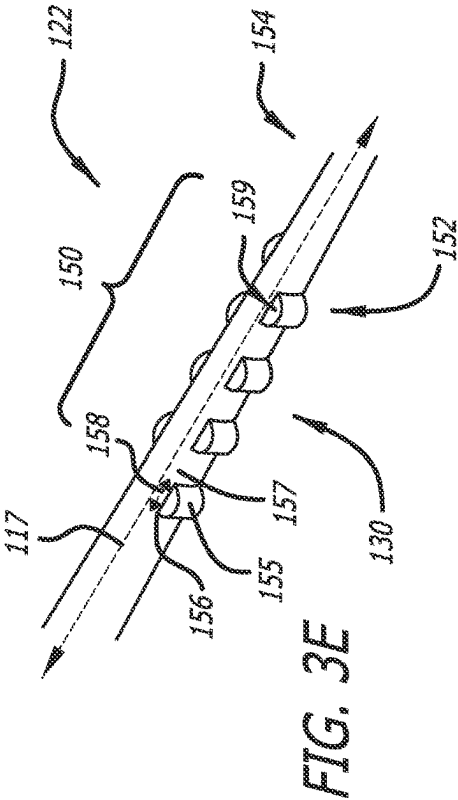
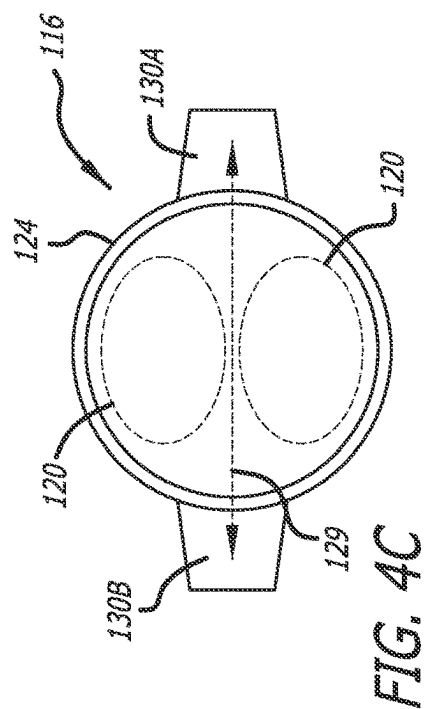
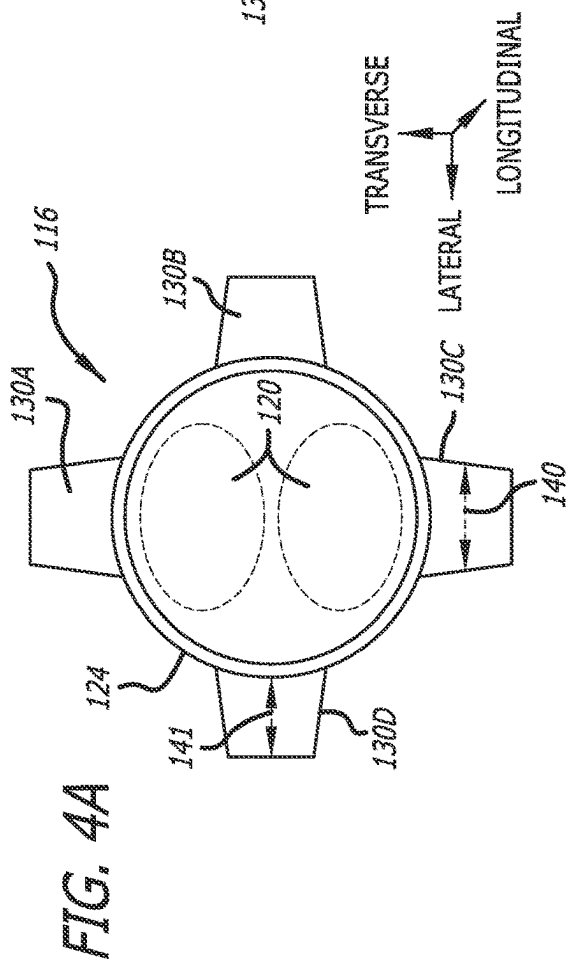
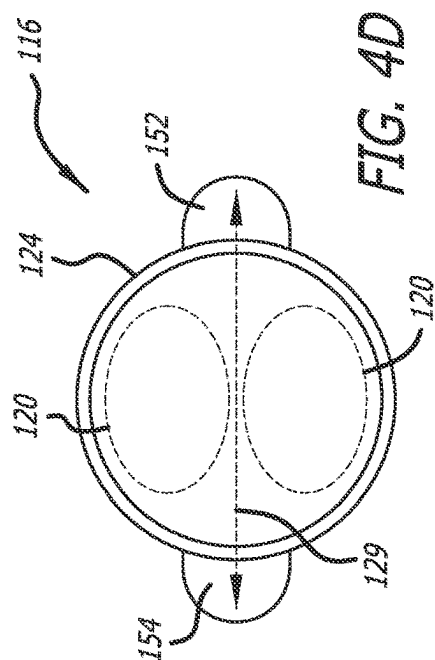
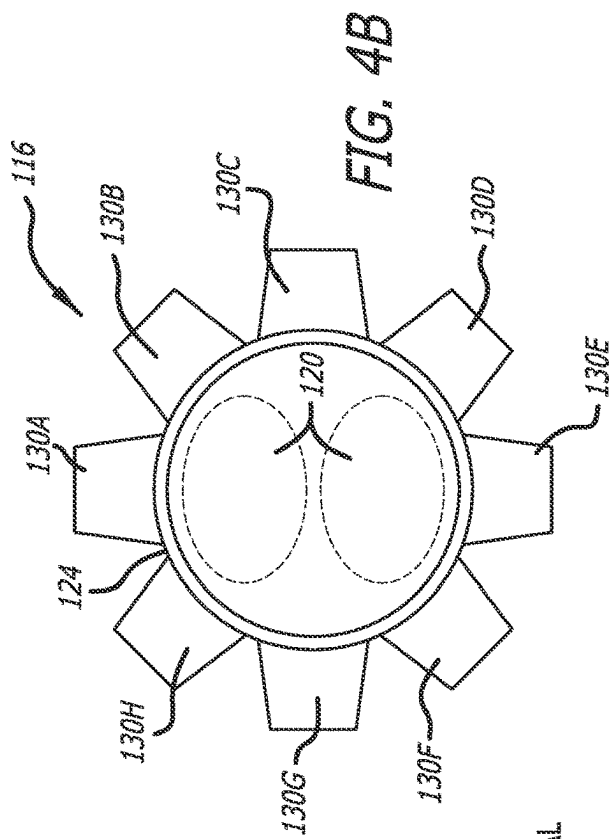
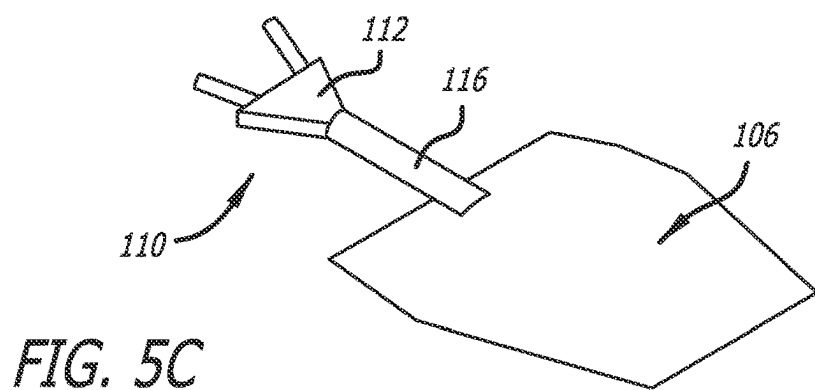
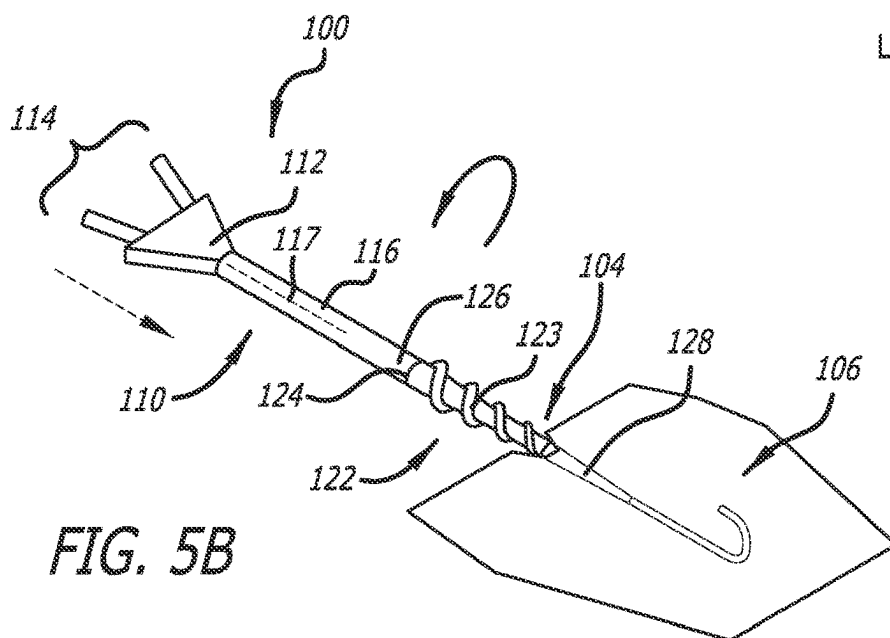
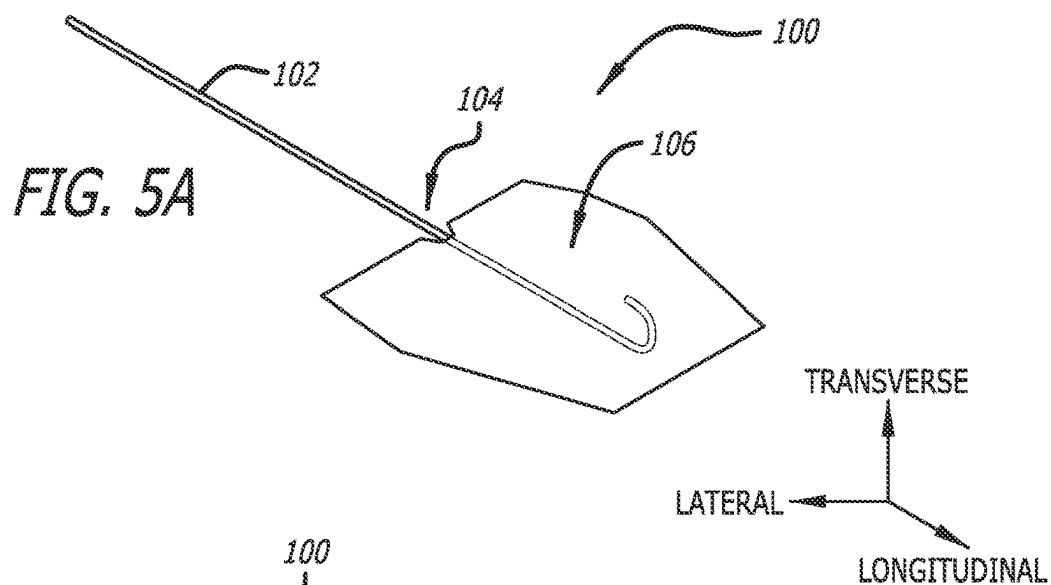


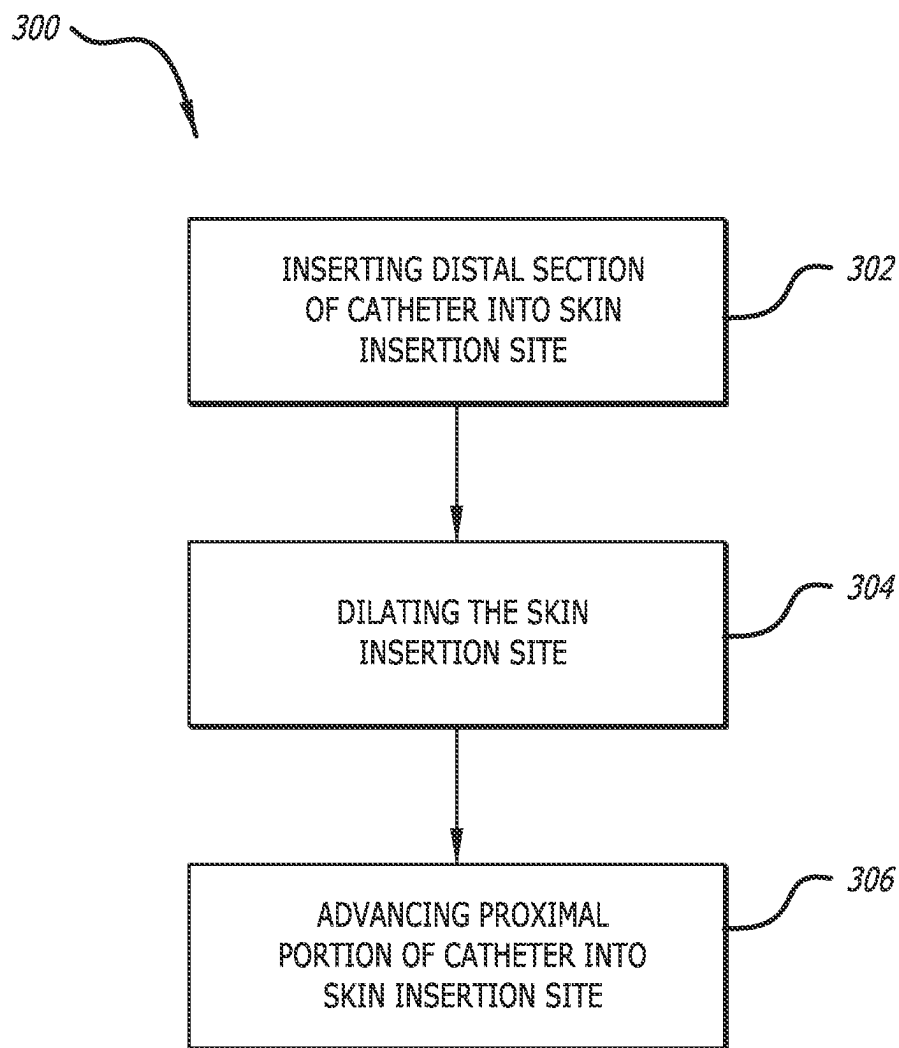
FIG 1









*FIG. 6*

1

PASSIVE DILATION IN CATHETER INSERTION SYSTEMS

PRIORITY

This application claims the benefit of priority to U.S. Provisional Application No. 63/121,761, filed Dec. 4, 2020, which is incorporated by reference in its entirety into this application.

BACKGROUND

In some catheter placement procedures, dilation of the tissue surrounding an insertion site is a core activity that occurs during catheter placement. Proper dilation of the tissue allows an improved fit of the catheter. Current dilation techniques are done by hand, using a separate hardened component and/or a skin nick to relieve the tension in the skin while dilating the tissue. These techniques require the insertion and removal of various instruments adding complexity, time and increasing the risk of infection. Furthermore, these techniques may lead to undesired tissue deflection and require increased pushing forces resulting in over dilation of the tissue surrounding the insertion site. It would be beneficial to be able to dilate only the tissue necessary for the fit of the catheter and reduce the necessity of the skin nick. Disclosed herein is a system and method of use that address the foregoing.

SUMMARY

Disclosed herein is a distal section of a catheter configured to enter a skin insertion site including a tapered junction including one or more dilation structures configured to dilate the skin insertion site, and a distal portion extending from a distal end of the tapered junction, the distal portion having a diameter smaller than a proximal portion of the catheter.

In some embodiments, the one or more dilation structures includes one or more protrusions or a helix extending from a side wall of the tapered junction.

In some embodiments, the one or more dilation structures extend radially from the side wall of the tapered junction.

In some embodiments, each protrusion includes a proximal edge and a distal edge.

In some embodiments, each protrusion extends longitudinally from the proximal edge to the distal edge.

In some embodiments, each protrusion extends laterally from the side wall to an apex.

In some embodiments, the apex is located between a longitudinal mid-point and one of the proximal edge or the distal edge, including the proximal edge or the distal edge.

In some embodiments, each protrusion defines a proximal face extending between the apex and the proximal edge.

In some embodiments, each protrusion defines a distal face extending between the apex and the distal edge.

In some embodiments, the proximal face defines a cutting edge extending longitudinally through a midline of the proximal face.

In some embodiments, the distal face defines a cutting edge extending longitudinally through a midline of the distal face.

In some embodiments, the protrusions are longitudinally organized into a first line of two or more protrusions and a second line of two or more protrusions disposed on opposite sides of the tapered junction.

2

In some embodiments, the proximal face and the distal face of each protrusion coordinate to define a rounded outer profile.

In some embodiments, each protrusion of the first line is longitudinally aligned with each protrusion of the second line.

In some embodiments, each protrusion of the first line is longitudinally offset from each protrusion of the second line.

In some embodiments, the helix includes a right handed helix or a left handed helix.

In some embodiments, the helix includes beveled or chamfered edges.

Also disclosed herein is a method of inserting a catheter including inserting a distal section of a catheter into a skin insertion site, the distal section having a tapered junction including one or more dilation structures configured to dilate the skin insertion site and a distal portion extending from a distal end of the tapered junction, the distal portion having a diameter smaller than a proximal portion of the catheter, dilating the skin insertion site, and advancing the proximal portion of the catheter into the skin insertion site.

In some embodiments, the one or more dilation structures include one or more protrusions or a helix extending from a side wall of the tapered junction.

In some embodiments, each protrusion includes a proximal face and a distal face.

In some embodiments, the proximal face defines a cutting edge extending longitudinally through a midline of the proximal face.

In some embodiments, the distal face defines a cutting edge extending longitudinally through a midline of the distal face.

In some embodiments, dilating the skin insertion site includes using the cutting edge to dilate the insertion site.

In some embodiments, dilating the skin insertion site includes rotating a helical dilation structure about a longitudinal axis to dilate the insertion site.

In some embodiments, wherein the helix includes a right handed helix or left handed helix.

In some embodiments, the helix includes beveled or chamfered edges.

In some embodiments, dilating the skin insertion site includes using the beveled or chamfered edges to dilate the insertion site.

In some embodiments, dilating the skin insertion site includes twisting the distal section of the catheter around a longitudinal axis, clockwise or counterclockwise.

In some embodiments, dilating the skin insertion site includes wiggling the distal section of the catheter about a transverse or lateral axis.

These and other features of the concepts provided herein will become more apparent to those of skill in the art in view of the accompanying drawings and following description, which describe particular embodiments of such concepts in greater detail.

DRAWINGS

A more particular description of the present disclosure will be rendered by reference to specific embodiments thereof that are illustrated in the appended drawings. It is appreciated that these drawings depict only typical embodiments of the invention and are therefore not to be considered limiting of its scope. Example embodiments of the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

FIG. 1 illustrates a plan view of a catheter insertion system, in accordance with some embodiments.

FIG. 2 illustrates a perspective view of components of a catheter insertion system including a tapered junction, in accordance with some embodiments.

FIG. 3A illustrates a schematic view of an exemplary dilation structure, in accordance with some embodiments.

FIGS. 3B-3I illustrates perspective views of various embodiments of the one or more dilation structures, in accordance with some embodiments.

FIGS. 4A-4D illustrate cross sectional views of various embodiments of the dilation structures, in accordance with some embodiments.

FIGS. 5A-5C illustrate an exemplary method of inserting a catheter into an insertion site, in accordance with some embodiments.

FIG. 6 illustrates a flow chart of an exemplary method of inserting a catheter, in accordance with some embodiments.

DESCRIPTION

Before some particular embodiments are disclosed in greater detail, it should be understood that the particular embodiments disclosed herein do not limit the scope of the concepts provided herein. It should also be understood that a particular embodiment disclosed herein can have features that can be readily separated from the particular embodiment and optionally combined with or substituted for features of any of a number of other embodiments disclosed herein.

Regarding terms used herein, it should also be understood the terms are for the purpose of describing some particular embodiments, and the terms do not limit the scope of the concepts provided herein. Ordinal numbers (e.g., first, second, third, etc.) are generally used to distinguish or identify different features or steps in a group of features or steps, and do not supply a serial or numerical limitation. For example, “first,” “second,” and “third” features or steps need not necessarily appear in that order, and the particular embodiments including such features or steps need not necessarily be limited to the three features or steps. Labels such as “left,” “right,” “top,” “bottom,” “front,” “back,” and the like are used for convenience and are not intended to imply, for example, any particular fixed location, orientation, or direction. Instead, such labels are used to reflect, for example, relative location, orientation, or directions. Singular forms of “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

With respect to “proximal,” a “proximal portion” or a “proximal-end portion” of, for example, a catheter disclosed herein includes a portion of the catheter intended to be near a clinician when the catheter is used on a patient. Likewise, a “proximal length” of, for example, the catheter includes a length of the catheter intended to be near the clinician when the catheter is used on the patient. A “proximal end” of, for example, the catheter includes an end of the catheter intended to be near the clinician when the catheter is used on the patient. The proximal portion, the proximal-end portion, or the proximal length of the catheter can include the proximal end of the catheter; however, the proximal portion, the proximal-end portion, or the proximal length of the catheter need not include the proximal end of the catheter. That is, unless context suggests otherwise, the proximal portion, the proximal-end portion, or the proximal length of the catheter is not a terminal portion or terminal length of the catheter.

With respect to “distal,” a “distal portion” or a “distal-end portion” of, for example, a catheter disclosed herein includes

a portion of the catheter intended to be near or in a patient when the catheter is used on the patient. Likewise, a “distal length” of, for example, the catheter includes a length of the catheter intended to be near or in the patient when the catheter is used on the patient. A “distal end” of, for example, the catheter includes an end of the catheter intended to be near or in the patient when the catheter is used on the patient. The distal portion, the distal-end portion, or the distal length of the catheter can include the distal end of the catheter; however, the distal portion, the distal-end portion, or the distal length of the catheter need not include the distal end of the catheter. That is, unless context suggests otherwise, the distal portion, the distal-end portion, or the distal length of the catheter is not a terminal portion or terminal length of the catheter.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by those of ordinary skill in the art.

FIG. 1 illustrates a plan view of a catheter insertion system **100**, in accordance with some embodiments. In some embodiments, the catheter insertion system **100** includes a catheter **110** having a hub **112** disposed at a proximal end and including one or more extension legs **114** extending proximally from the proximal end of the hub **112**. In some embodiments, the catheter **110** includes a catheter tube **116** distally extending from the distal end of the hub **112**. In some embodiments, the catheter tube **116** may be configured to slide over a guidewire **102**. The catheter tube **116** includes a proximal end **119** and a distal section **121** terminating in a distal tip **118** and can define one or more catheter lumens **120** therein. In some embodiments, the catheter **110** may include a Rapid Insertion Central Catheter (“RICC”), such as described, for example, in U.S. Pat. Nos. 10,376,675; 11,517,719; U.S. Publication No. 2021/0069471; U.S. Publication No. 2021/0113809; U.S. Publication No. 2021/0113810; and U.S. Publication No. 2021/0121661, each of which is incorporated by reference in its entirety into this application. As used herein, the catheter insertion system **100** can be used to place a RICC to access the vasculature of a patient. However, it will be appreciated that embodiments disclosed herein can be used to place various catheters, cannulas, single lumen catheters, multi-lumen catheters, intravenous (IV) catheters, peripheral intravenous line (PIV) catheters, peripherally inserted central catheters (PICC), central venous catheter (CVC), dialysis catheters, drainage catheters, and the like, without limitation.

Placing a catheter **110** requires dilating the tissue surrounding an insertion site since typically the insertion site is smaller than the diameter of the catheter **110**. Current dilation techniques use a hardened component that can cause tissue deflection and excessive stretching of the skin tissue as the hardened component is forced through the insertion site. The less movement of the insertion site and the less tissue dilated around the insertion site can lead to a more accurate placement of the catheter **110** within the body and a better fit of the catheter **110** within the insertion site.

In some embodiments, a cross-sectional shape of the catheter tube **116** may be cylindrical, a triangular prism, a rectangular prism, or the like, although other cross-sectional shapes are also contemplated. In some embodiments, the catheter tube **116** may be configured to have a tapered junction **122** between the proximal end **119** of the catheter tube **116** and the distal section **121** of the catheter tube **116**. In some embodiments, the tapered junction **122** may be configured to include one or more dilation structures **123** extending from a side wall **124** of the tapered junction **122** that will be described in more detail herein.

5

FIG. 2 illustrates a perspective view of components of a catheter insertion system 100 including the tapered junction 122, in accordance with some embodiments. In some embodiments, the catheter tube 116 includes a proximal portion 125, the tapered junction 122 having the side wall 124 and a distal portion 127 distal the tapered junction 122. In some embodiments, the catheter tube 116 defines a central longitudinal axis 117. The proximal portion 125 has a second diameter 126 and the distal portion 127 has a first diameter 128. In some embodiments, the second diameter 126 is larger than the first diameter 128. The tapered junction 122 is a portion of the catheter tube 116 that is tapered from the proximal portion 125 to the distal portion 127 and from the second diameter 126 to the first diameter 128. In some embodiments, the tapered junction 122 may be evenly tapered, or unevenly tapered from the proximal portion 125 to the distal portion 127. In some embodiments, the first diameter 128 and the second diameter 126 may be associated with the French catheter scale. In some embodiments, the tapered junction 122 may be configured to include one or more dilation structures 123 that may be configured to have a specific shape to dilate the tissue surrounding an insertion site that will be described in more detail herein.

FIG. 3A illustrates a schematic view of an exemplary dilation structure, in accordance with some embodiments. In some embodiments, the one or more dilation structures 123 may be configured to include structures that extend from the tapered junction 122 that dilate tissues upon insertion of the catheter 110 into an insertion site. In some embodiments, the one or more dilation structures 123 include a protrusion 130 extending from the side wall 124. In some embodiments, the one or more dilation structures extend radially from the side wall 124 or longitudinally from a proximal edge 135 to a distal edge 137. In some embodiments, the protrusion 130 may be configured to extend from the side wall 124 to an apex 132. In some embodiments, the apex 132 defines the furthest point of the protrusion 130 from the side wall 124. In some embodiments, the protrusion 130 further defines a proximal face 134, extending laterally between the apex 132 and the proximal edge 135 and a distal face 136 extending between the apex 132 and the distal edge 137. In some embodiments, the protrusion 130 includes a length 138 and a width 140. In some embodiments, as illustrated in FIGS. 3C-3D, the length 138 may remain the same while the apex 132 may be disposed anywhere along the length 138 including a longitudinal mid-point, towards and including the proximal edge 135 or the distal edge 137.

In some embodiments, the one or more dilation structures 123 may include one or more protrusions 130 or helices 160 that will be described in more detail herein. FIGS. 3B-3I illustrate perspective views of various embodiments of the protrusions 130 or helices 160, in accordance with some embodiments. As will be appreciated, the protrusions 130 can extend radially from the side wall 124. In some embodiments, the one or more protrusions 130 can extend from the side wall 124 at a same distance or at a different distance from the side wall 124 of the tapered junction 122, as will be described in more detail herein. In some embodiments, as illustrated in FIGS. 3B-3C, the protrusion 130 can include various geometries. For example, as illustrated in FIG. 3B, the proximal face 134A or distal face 136A can define a cutting edge extending longitudinally through a midline 139 of the proximal face 134A or the distal face 136A. In some embodiments, the cutting edge may be configured to assist the user with dilating the tissue around the insertion site and avoid tissue deflection. As illustrated in FIG. 3D, the tapered junction 122 can include a plurality of protrusions 130A-C,

6

defining the longitudinally rounded distal face 136, the midline 139 and the longitudinally rounded proximal face 134. An apex 132A of a first protrusion 130A can be longitudinally offset from an apex 132B of a second protrusion 130B.

FIG. 3E illustrates a perspective view of the one or more protrusions 130 wherein the one or more protrusions 130 include a plurality of ribs 150, wherein the plurality of ribs 150 are repeated rounded protrusions. In some embodiments, each rib of the plurality of ribs 150 includes a rib face 155 extending from a proximal edge 156 to a distal edge 157, a rib length 158 extending from the proximal edge 156 to the distal edge 157, and a rib width 159. In some embodiments, the rib face 155 defines the rounded outer profile of the protrusion 130. In some embodiments, the rib length 158 of the plurality of ribs 150 may be the same or varied. In some embodiments, the rib width 159 of the plurality of ribs 150 may be the same or varied. In an embodiment, the rib width 159 tapers from a larger rib width 159 proximally to a smaller rib width 159 distally.

In some embodiments, the plurality of ribs 150 may be randomly distributed or evenly distributed on the side wall 124 of the tapered junction 122. In some embodiments, the plurality of ribs 150 may be longitudinally organized into a first line 152 of two or more ribs and a second line 154 of two or more ribs. In an embodiment, each rib in the first line 152 may be longitudinally aligned with each rib in the second line 154. In an embodiment as illustrated in FIG. 3E, each rib in the first line 152 may be longitudinally offset from each rib in the second line 154. Each rib in the first line 152 may be evenly spaced or unevenly spaced from the other ribs in the first line 152 and each rib in the second line may be evenly spaced or unevenly spaced from the other ribs in the second line. The first line 152 and the second line 154 can be disposed on opposite sides of the tapered junction 122. In some embodiments, the first line 152 may be aligned along a lateral side and the second line 154 may be aligned along a transverse side, or at an angle therebetween.

In an embodiment, the first line of ribs 152 can be arranged in a helical pattern about the longitudinal axis 117, where each rib of the first line of ribs 152 are rotationally offset by an angle about the longitudinal axis 117 and longitudinally offset by a distance. In an embodiment, each rib of the first line of ribs 152 can be rotationally offset by the same angle or by different angles. In an embodiment, each rib of the first line of ribs 152 can be longitudinally offset by the same distance or different distances.

FIGS. 3F-3I illustrates a perspective view of the one or more dilation structures 123 wherein the dilation structure 123 includes the helix 160 extending radially about longitudinal axis 117 from the side wall 124. In some embodiments, the helix 160 may be a right handed helix or a left handed helix. In some embodiments, as illustrated in FIG. 3F, the helix 160 may have a longitudinal length 161. In some embodiments, e.g. FIG. 3F, the dilation structure 123 can include a relatively high number of revolutions per longitudinal length 161, providing a relatively "tighter" helical structure. In some embodiments, e.g. FIG. 3G, the dilation structure 123 can include a relatively low number of revolutions per longitudinal length 161, providing a relatively "looser" helical structure.

In some embodiments, each revolution of the helix 160 may extend radially the same distance from the side wall 124, as illustrated in FIG. 3G, or each revolution may extend different distances from the side wall 124. For example, as illustrated in FIG. 3H, the helix 160 may extend a greater distance from the side wall 124 at a proximal end and may

extend a lesser distance from the side wall **124** at a distal end. Advantageously, by rotating the tapered junction **122**, including the helical dilation structure **123**, the mechanical action of the helix **160** pulls the tapered junction **122** through the insertion site **104**, avoiding tissue deflection or distortion. In some embodiments, as illustrated in FIG. 3I, the helix **160** may be configured to have beveled, chamfered, squared or rounded edges. The beveled or chamfered edges may be configured to partially cut or fully cut the tissue surrounding the insertion site **104** as the helix **160** is rotated, further dilating the tissue. Although FIGS. 3A-3I illustrate various embodiments of the one or more dilation structures **123** of the tapered junction **122**, other shapes or protrusions are considered.

FIGS. 4A-4D illustrate cross sections of the one or more dilation structures **123** including the one or more protrusions **130** of the tapered junction **122**, in accordance with some embodiments. In some embodiments, the catheter tube **116** includes the one or more lumens **120** extending there-through. In some embodiments, the protrusions **130** include a height **141**. As used herein, the “height” of the protrusion **130** is a distance of an outer most point of the protrusion **130** extending radially from the side wall **124**. In some embodiments, the protrusion **130** defines a width **140**. As used herein a “width” is a distance extending parallel to the side wall **124**. In some embodiments, the protrusions **130** may have the same width **140** and the same height **141** or different widths **140** and different heights **141**.

As illustrated in FIG. 4A, the tapered junction **122** may include four protrusions **130A-D** extending radially from the side wall **124** of the tapered junction **122**. In some embodiments, a first ridge **130A** may be equidistant from a second ridge **130B** and a fourth ridge **130D**. In some embodiments, the fourth ridge **130D** may be equidistant from the first ridge **130A** and a third ridge **130C**. Simply put, the four protrusions **130A-D** may be evenly spaced radially around the tapered junction **122**. As illustrated in FIG. 4B, in some embodiments, the tapered junction **122** may include eight ridges **130A-H** extending from the side wall **124** of the tapered junction **122**. In some embodiments, the eight ridges **132A-H** may be even spaced, unevenly spaced or clustered together around the tapered junction **122**. Although eight ridges are illustrated, greater or lesser number of ridges are also contemplated.

As illustrated in FIG. 4C, in some embodiments, the tapered junction **122** may include two protrusions **130A**, **130B** extending from the side wall **124** of the catheter tube **116**. In some embodiments, the first protrusion **130A** and the second protrusion **130B** define a lateral axis **129**. In some embodiments, the first protrusion **130A** may be longitudinally aligned with the second protrusion **130B**, defining a lateral axis **129** on opposite sides of the tapered junction **122**. In some embodiments, the first protrusion **130A** may be longitudinally offset from the second protrusion **130B**. In some embodiments, the one or more protrusions **130** may include the plurality of ribs **150**. In some embodiments, the plurality of ribs **150** may be organized in the first line **152** and the second line **154**. As illustrated in FIG. 4D, in some embodiments, the first line **152** may longitudinally aligned with the second line **154**, on opposite sides of the catheter tube **116**. In some embodiments, the first line **152** may be longitudinally offset from the second line **154**. In some embodiments, the plurality of ribs **150** may radially extend evenly or unevenly from the side wall **124** to define different distances from the side wall **124** of the tapered junction **122**. In some embodiments, the plurality of ribs **150** located proximally may radially extend a greater distance from the

side wall **124** than the ribs located distally. In some embodiments, the plurality of ribs **150** may be organized into a plurality of lines, evenly spaced around the tapered junction **122**. In some embodiments, the plurality of ribs **150** may be spaced in an organized configuration or a random configuration around the catheter tube **116**. In some embodiments the plurality of ribs may be arranged in a helical line about the longitudinal axis **117**.

FIGS. 5A-5C illustrate an exemplary method of using inserting a catheter **110** into an insertion site **104**. In some embodiments, the catheter insertion system **100** includes a guidewire **102**. As illustrated in FIG. 5A, the guidewire **102** may be inserted into an insertion site **104** in a target area **106**. In some embodiments, the target area **106** may include a target vessel. In some embodiments, a user may confirm the guidewire **102** is in the correct target location. In some embodiments, as illustrated in FIG. 5B, the catheter insertion system **100** further includes the catheter **110** having the hub **112** disposed at a proximal end and including one or more extension legs **114** extending proximally from the proximal end of the hub **112**. The catheter **110** includes the catheter tube **116** distally extending from the distal end of the hub **112**. The catheter tube **116** may be configured to slide over the guidewire **102**. The catheter tube **116** includes the proximal end **119** and a distal section **121**, terminating in the distal tip **118** and can define the one or more catheter lumens **120** therein. The catheter tube **116** further includes the proximal portion **125**, the tapered junction **122** having the side wall **124** and the distal portion **127**. The proximal portion **125** has the second diameter **126** and the distal portion **127** has the first diameter **128** with the second diameter **126** being larger than the first diameter **128**. The tapered junction **122** may be configured to include one or more dilation structures **123** radially extending from the side wall **124** and configured to have a specific shape to dilate the tissue surrounding the skin insertion site **104** while mitigating tissue deflection. The dilation structures **123** can include one or more protrusions **130** or helices **160**.

Once the guidewire **102** is placed at the correct target location, the catheter **110** may be slid over the guidewire **102** and urged into the skin insertion site **104** as illustrated in FIG. 5B. Since the distal portion **127** defines a smaller first diameter **128**, the distal portion **127** can enter the skin insertion site **104** more easily. The tapered junction **122** including the one or more dilation structures **123** can then dilate the skin insertion site **104** for the larger second diameter **126** and the proximal portion **125**. As the catheter **110** is urged into the skin insertion site **104**, the one or more dilation structures **123** facilitates dilating the tissue around the skin insertion site **104** by stretching the tissue around the skin insertion site **104** along a lateral axis. For example, a rotating helix **160** may be configured to use mechanical advantage to pull the tapered junction **122** through the skin insertion site **104**, minimizing tissue deflection. In some embodiments, the user may further twist the catheter **110** clockwise or counterclockwise about the longitudinal axis **117** while urging the catheter **110** into the skin insertion site **104** to further dilate the tissue surrounding the insertion site **104**.

In some embodiments, the user may “wiggle” the catheter **110** back and forth along the lateral axis, transverse axis or a combination thereof, while urging the catheter **110** into the insertion site **104** to further dilate the tissue surrounding the insertion site **104**. In some embodiments, e.g. FIG. 3D, the user may wiggle the catheter **110** back and forth along the lateral or a transverse axis. In some embodiments, e.g. FIGS. 3E-3I, the user may wiggle and twist the catheter **110** while

urging the catheter **110** into the insertion site **104**. In some embodiments, the user may continue to wiggle or twist the catheter **110** until the catheter **110** is completely inserted into the insertion site **104**, as illustrated in FIG. 5C. Once the catheter **110** is fully inserted, the user may confirm proper target location placement of the catheter **110** through imaging. In some embodiments, using the catheter **110** to dilate the insertion site **104** leads to an increased fit of the catheter **110** into the insertion site **104**, while minimizing tissue deflection around the insertion site **104**.

FIG. 6 illustrates a flow chart of an exemplary method **300** of inserting a catheter **110** into an insertion site **104** using a catheter insertion system **100**, in accordance with some embodiments. In some embodiments, the method **300** includes inserting a distal section **121** of the catheter **110** into the insertion site **104** (block **302**). In some embodiments, the distal section **121** includes the tapered junction **122** having one or more dilation structures **123** radially extending from the side wall **124** and a distal portion **127** extending from the distal end of the tapered junction, having the first diameter **128**.

The method **300** includes dilating the skin insertion site **104** (block **304**). In some embodiments, dilating the insertion site **104** includes dilating from the smaller first diameter **128** to the larger second diameter **126**. In some embodiments, dilating the skin insertion site **104** includes urging the tapered junction **122** of the catheter **110** into the skin insertion site **104**, the tapered junction **122** includes the one or more dilation structures **123** radially extending from the side wall **124**. In some embodiments, the one or more dilation structures **123** include the one or more protrusions **130** or the helix **160**. In some embodiments, dilating the insertion site **104** includes twisting the catheter **110** clockwise or counterclockwise around the central longitudinal axis **117**. In some embodiments, dilating the insertion site **104** includes wiggling the catheter **110** along the transverse or lateral axis.

The method **300** further includes advancing the proximal portion **125** of the catheter **110** into the insertion site **104** (block **306**). In some embodiments, the proximal portion **125** includes the second diameter **126**. In some embodiments, the second diameter **126** is greater than the first diameter **128**.

While some particular embodiments have been disclosed herein, and while the particular embodiments have been disclosed in some detail, it is not the intention for the particular embodiments to limit the scope of the concepts provided herein. Additional adaptations and/or modifications can appear to those of ordinary skill in the art, and, in broader aspects, these adaptations and/or modifications are encompassed as well. Accordingly, departures may be made from the particular embodiments disclosed herein without departing from the scope of the concepts provided herein.

What is claimed is:

1. A catheter configured to enter a skin insertion site, comprising:

- a hub disposed proximally and including one or more extension legs extending proximally therefrom;
- a catheter tube extending distally from the hub and defining a first catheter lumen and a second catheter lumen, each of which are in fluid communication with an extension leg of the one or more extension legs;

a tapered junction coupled to a distal end of the catheter tube and including a dilation structure configured to dilate the skin insertion site, the dilation structure comprising:

- a first protrusion extending along a longitudinal axis between a first distal edge and a first proximal edge and extending along a lateral axis to a first apex, the first apex disposed proximate a first mid-point between the first distal edge and the first proximal edge, the first protrusion including a first cutting edge; and
- a second protrusion extending along the longitudinal axis between a second distal edge and a second proximal edge and extending along the lateral axis to a second apex, the second apex disposed proximate a second mid-point between the second distal edge and the second proximal edge, the second protrusion including a second cutting edge; and
- a distal portion extending from a distal end of the tapered junction, the distal portion having a diameter smaller than a diameter of a proximal portion of the catheter tube and defining a distal part of the first catheter lumen, the second catheter lumen terminating at an aperture disposed in a side wall and disposed adjacent the distal end of the catheter tube.

2. The catheter according to claim 1, wherein an outermost lateral diameter extending between the first apex and the second apex, is greater than, the diameter of the catheter tube.

3. The catheter according to claim 2, wherein an outermost transverse diameter of the dilation structure, extending perpendicular to the lateral axis is less than the diameter of the catheter tube.

4. The catheter according to claim 1, wherein the first protrusion defines a first proximal face extending between the first apex and the first proximal edge, and the second protrusion defines a second proximal face extending between the second apex and the second proximal edge.

5. The catheter according to claim 4, wherein the first protrusion defines a first distal face extending between the first apex and the first distal edge, and the second protrusion defines a second distal face extending between the second apex and the second distal edge.

6. The catheter according to claim 4, wherein the first proximal face defines the first cutting edge extending longitudinally through a first midline of the first proximal face, and the second proximal face defines the second cutting edge extending longitudinally through a second midline of the second proximal face.

7. The catheter according to claim 5, wherein the first distal face defines the first cutting edge extending longitudinally through a first midline of the first distal face, and the second distal face defines the second cutting edge extending longitudinally through a second midline of the second distal face.

8. The catheter according to claim 1, wherein the tapered junction, the distal portion and the distal end of the catheter tube are configured to be placed subcutaneously through the skin insertion site.

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